

TSS DFIR REPORT >> Vantara Financial Group

June 5th Incident Digital Forensics Report

DISCLAIMER: All entities mentioned in this document are completely fictional and have no connection to any real-world entity. This document also isn't actually TLP:AMBER – it's just there to make the report look more realistic. This document is more like TLP:CLEAR.

Table of Contents

Executive Summary..... 3

Investigation Scope & Background..... 4

Incident Timeline & Technical Analysis..... 5

Recommendations..... 11

Appendix and IoCs..... 12

Executive Summary

Incident Overview & Timeline

On June 5th, 2026, an attacker gained unauthorized access to the network of Vantara Financial Group, compromising endpoints as they moved throughout it. Evidence suggests that during the attack, the attacker used a compromised internal endpoint to drop a malicious file onto an in-scope workstation, which the attacker subsequently executed. With control over the workstation, the attacker deployed two persistence mechanisms and accessed a document that contained the company's Q1 2026 revenue summary. The attacker staged the document and archived its contents, but it cannot be confirmed if the attacker exfiltrated the archive. Towards the end of the attacker's activity on the endpoint, the attacker attempted to move beyond the compromised workstation and into the company's file servers.

However, given the limited scope of the investigation and being confined to one endpoint, TSS was unable to pinpoint the root cause of the network compromise and subsequent actions the attacker performed on the network.

Impact

Due to the limited scope of the investigation, TSS cannot evaluate the total impact of the compromise.

Investigation Scope & Background

Incident Authorization and Background

On June 5th, 2026, Vantara Financial Group's internal SOC engaged TSS's forensics team after receiving alerts of anomalous authentication activity and exfiltration activity in a finance department fileserver.

Investigation Purpose

The purpose of the investigation is to determine the scope of the breach, its impact, and the optimal approach to removing the attacker from the endpoint. Other goals of the investigation included determining the method of initial access, the establishment of persistence mechanisms, attempts of lateral movement, and extent of exfiltration.

Scope

The investigation scope included a single endpoint in the Vantara Financial Group network – **VFG-CTR-W019**. The evidence provided from the endpoint was a KAPE (Kroll Artifact Parser and Extractor) extraction containing various artifacts, such as Windows event logs, NTFS metadata, registry hives, and other miscellaneous items.

However, the rest of the Vantara Financial Group's network was out of scope, leaving findings unavailable for endpoints outside of the one specified in scope. Additionally, the compromised user's **NTUSER.DAT** was missing from the provided triage capture.

Incident Timeline & Technical Analysis

Below is a timeline of the attack against **VFG-CTR-W019**. The corresponding MITRE ATT&CK technique is provided in each stage. All timestamps provided are in UTC.

1. Lateral Movement (T1021.002, T1570, T1021.001)

At the start of the attack, the attacker dropped a malicious file into **VFG-CTR-W019** as part of a lateral movement attempt.

The logon patterns and file creation dates on the compromised endpoint suggest that on **2026-06-05 5:56:46 UTC**, the internal IP address **192.168.215.227** dropped a malicious file (**C:\USERS\DANIEL.AVERY\DOWNLOADS\VPNSETUP_V2.1.EXE**) onto **VFG-CTR-W019** using SMB.

This action was observed through the endpoint's Security event logs and the NTFS Master File Table (MFT).

UTC Timestamp	Event	Source
2026-06-05 5:56:46	Logon type 3 (Network) observed from internal IP address 192.168.215.227 – Logon ID 0x3BAD5	Security event log
2026-06-05 5:57:32	File VPNSetup_v2.1.exe was created in C:\Users\daniel.avery\downloads\	MFT
2026-06-05 5:57:59	Logoff type 3 observed from internal IP address 192.168.215.227 – Logon ID 0x3BAD5	Security event log

However, it must be noted that due to the absence of web browser artifacts, email artifacts, and network captures in the evidence pool, the investigation could not evaluate if the file came from a web download. SMB remains the more likely option due to the event patterns and the absence of a Zone.Identifier Alternate Data Stream (ADS), which would mark files downloaded via web browsers or email clients.

After dropping the malicious file, the attacker logged into the account **daniel.avery** on **VFG-CTR-W019** via Remote Desktop Protocol (RDP) and executed the dropped file. This logon deviated from typical sign-in behavior from **192.168.215.227** – according to the Security event logs, there have not been any recorded logon events from **daniel.avery**

until the attack. This anomalous logon event was observed through the endpoint's Security event logs.

UTC Timestamp	Event	Source
2026-06-05 5:58:15	Logon type 10 (RDP) into account daniel.avery from internal IP address 192.168.215.227 - Logon ID to track: 0x885A6	Security event log

2. Execution & Staged Payload (TA0002 & T1105)

After gaining access to the account **daniel.avery** on **VFG-CTR-W019**, the attacker executed the file earlier dropped via SMB (**VPNSetup_v2.1.exe**) at **2026-06-05 5:58:41**. The file then dropped a second stage payload – named **SVCHOSTS.EXE** – into the user's temporary folder (**C:\USERS\DANIEL.AVERY\APPDATA\LOCAL\TEMP**). This file attempts to masquerade (T1036.005) as the legitimate Windows binary – **svchost.exe** – that is typically found in the System32 folder.

These actions can be observed through the endpoint's Prefetch files and Master File Table. Please note the timestamp deviation (Marked in red text).

UTC Timestamp	Event	Source
2026-06-05 5:58:41	VPNSetup_v2.1.exe was executed	Prefetch Files (.pf)
2026-06-05 5:36:12	SVCHOSTS.EXE was created in C:\USERS\DANIEL.AVERY\APPDATA\LOCAL\TEMP\	MFT
2026-06-05 6:01:48	SVCHOSTS.EXE was executed	Prefetch files (.pf)

In the table seen above, the event corresponding to the creation of **SVCHOSTS.EXE** was moved in front of the execution time of **VPNSetup_v2.1.exe**. This is for three primary reasons:

1. 5:36 UTC is an abnormal timestamp for the time range of this attack. If **SVCHOSTS.EXE** was really created at this time, the attack timeline would be pushed back over 20 minutes earlier. However, there is no observable in-scope evidence that points to suspicious activity before 5:36.

2. According to the endpoint's Prefetch files, `VPNSetup_v2.1.exe` was executed prior to `SVCHOSTS.EXE`'s execution. However, the inverse should be the case if `SVCHOSTS.EXE` was created before `VPNSetup_v2.1.exe`.
3. There is an observed moment on the system around these timestamps that suggest the system's reported time travelled backwards over 20 minutes. This moment can be observed in the Windows Time Service operational logs, between two events. When the system time is changed, the event log also keeps track of the tick count, which is the number of milliseconds that have passed since the computer started. This is much harder to change compared to the system time.

UTC Timestamp	Tick Count	Event Name	New Time	Old Time
2026-06-05 05:49:36	2492218	TMP_OPS_SET_TIME	2026-06-05 05:49:36.821	2026-04-21 09:00:13.737
2026-06-05 05:37:18	2866359	TMP_OPS_SET_TIME	2026-06-05 05:37:18.923	2026-06-05 05:37:18.914

Between the first event and the second event, the tick count visibly increases by over 350,000 milliseconds (~6 minutes). However, between the two events, the UTC timestamp moves backwards by around 12 minutes. This signifies that the system's time moved backwards at some point. `SVCHOSTS.EXE` was likely created near this moment, which explains the skewed creation timestamp.

3. Persistence & Discovery (T1136, T1053.005 & TA0007)

Once `C:\USERS\DANIEL.AVERY\DOWNLOADS\VPNSETUP_V2.1.EXE` was executed, a malicious scheduled task named `MicrosoftEdgeUpdateCore` was created on `2026-06-05 6:01:16` by user `daniel.avery`. This scheduled task executed the binary `C:\Users\daniel.avery\AppData\Local\Temp\svchosts.exe` on logon. As seen in its name, the scheduled task attempted to masquerade (T1036.004) as a legitimate update service for the Microsoft Edge browser. This event can be found in the Task Scheduler Operational event logs, as event ID 106.

Afterwards, the attacker used PowerShell to run discovery commands against the compromised endpoint. Five discovery commands can be found in the PowerShell `ConsoleHost_history.txt` file. Correlating this information with the endpoint's Prefetch files also provides the timestamps for these commands.

The table can be found on the next page.

UTC Timestamp	Event	ATT&CK Mapping
2026-06-05 6:02:05	daniel.avery executes Powershell discovery command <code>whoami /all</code>	T1033
2026-06-05 6:02:11	daniel.avery executes Powershell discovery command <code>net user /domain</code>	T1087.002
2026-06-05 6:02:18	daniel.avery enumerates a security-enabled local group membership was enumerated (Builtin\Administrators (S-1-5-32-544)) via the execution of Powershell command <code>net localgroup administrators</code>	T1069.001
2026-06-05 6:02:25	daniel.avery executes Powershell discovery command <code>ipconfig /all</code>	T1016
2026-06-05 6:02:32	daniel.avery executes Powershell discovery command <code>nltest /domain_trusts</code>	T1482

As seen in the events, the attacker runs 5 discovery commands to learn more about the environment:

1. `whoami /all`, which displays security and identity details from the current signed-in user
2. `net user /domain`, which lists all user accounts in the Active Directory domain.
3. `net localgroup administrators`, which lists all user accounts in the local Administrators group
4. `ipconfig /all`, which lists detailed information about the computer's network configuration
5. `nltest /domain_trusts`, which lists all domain trusts for the current domain

Once the attacker ran the discovery commands, the attacker created a second persistence mechanism by creating a new user account named `helpdesk$` and giving it administrator privileges. The attacker attempted to masquerade the account as a machine account (T1036.010), as seen with the dollar sign suffix (\$) at the end of its name. The creation of the account can be observed through the Security event logs, under logon ID `0x885A6`.

These events can be observed in the next page.

UTC Timestamp	Event	Source
2026-06-05 6:02:59	daniel.avery, under logon ID 0x885A6, creates a new account named helpdesk\$.	Security event log
2026-06-05 6:02:59	The helpdesk\$ account is enabled	Security event log
2026-06-05 6:02:59	The password for helpdesk\$ is reset	Security event log
2026-06-05 6:02:59	helpdesk\$ is added to the users group	Security event log
2026-06-05 6:03:00	helpdesk\$ is added to the administrators group	Security event log

Other potential sources of exfiltration were also checked, and TSS confirmed that there was no persistence mechanisms deployed in:

1. Registry Run Keys
2. Services
3. Startup Folders
4. WMI Event Subscriptions

4. Collection & Exfiltration (T1560, T1074.001, T1005 & TA0010)

After setting up persistence mechanisms, the attacker collected a sensitive file, staging and compressing it into a ZIP archive named data_backup.zip, which was located in C:\Windows\Temp\. Upon extraction of resident data from the Master File Table, this ZIP archive was found to only contain one file – Q1_2026_Revenue_Summary.txt. The original location of this document is C:\Users\daniel.avery\Documents\VFG-Projects\FinanceReports\Q1_2026_Revenue_Summary.txt.

This can be observed in the Master File Table

UTC Timestamp	Event	Source
2026-06-05 6:04:29	Zip archive data_backup.zip is created as part of collection and staging.	Master File Table

As for data exposure, the attacker was observed to only access one document – Q1_2026_Revenue_Summary.txt. This can be seen with the endpoint's Automatic Destination files.

UTC Timestamp	Event	Source
2026-06-05 6:05:16	File Q1_2026_Revenue_Summary.txt, located in C:\Users\daniel.avery\Documents\VFG-Projects\FinanceReports\, was opened	Automatic Destinations

The system's shortcut files (LNK) were also examined, but they did not provide extra information.

The observed staging and collection activity heavily suggest the attacker exfiltrated the data, but this cannot be confirmed or detailed due to the scope of the investigation and the artifacts provided.

5. Lateral Movement (T1021.002)

During the attack, the attacker attempted lateral movement into one of Vantara Financial Group's file servers (VFG-FS01) using explicit credentials for the account svc.backup. This action can be observed in the PowerShell ConsoleHost_history.txt file.

UTC Timestamp	Event	Source
Timestamp unknown	daniel.avery attempts lateral movement via SMB using explicit credentials by executing Powershell command net use Z: \\VFG-FS01\Finance /user:VFG\svc.backup [PASS REDACTED]	PowerShell ConsoleHost History File

Note that this command contains cleartext passwords, which has been redacted in the report.

Additionally, note that the timestamp for this event is unavailable, as a corresponding timestamped event could not be located within the endpoint. Additionally, due to the scope of the investigation and the provided artifacts, whether the lateral movement attempt was successful or not is unconfirmed.

Recommendations

Although comprehensive recommendations cannot be provided due to the limited scope of the investigation, TSS recommends these immediate actions to contain the threat:

- If not done already, isolate the compromised host `VFG-CTR-W019` from the network.
- Delete the unauthorized user, `helpdesk$`.
- Remove the malicious scheduled task, `MicrosoftEdgeUpdateCore`.
- Force credential resets for accounts `daniel.avery` and `svc.backup`.
- Delete the malicious files on the system
 - o `SVCHOSTS.EXE`
 - o `VPNSetup_v2.1.exe`
- Continue investigation of other devices in the network, especially `192.168.215.227` and `VFG-FS01`

To prevent such compromises from happening in the future, TSS recommends the following:

- Improve auditing and logging across the network. This includes installing Sysmon on endpoints and enabling auditing for file shares.
- Improve the password policy. For example, this includes disallowing common password patterns, increasing length requirements, and allowing passphrases.

Appendix and IoCs

Appendix A: IoCs

Host from Lateral Movement: 192.168.215.227

Malicious Payload: VPNSETUP_V2.1.EXE SHA1:
d3c83599d21508c14ffe67d87abf5a83ddc4d157

Malicious Second Stage Payload:
C:\Users\daniel.avery\AppData\Local\Temp\svchosts.exe (SHA1:
Unavailable)

Backdoor User: helpdesk\$

Malicious scheduled task: MicrosoftEdgeUpdateCore

Staged Exfiltration Archive: C:\Windows\Temp\data_backup.zip

Appendix B: Extra Links and Information

Spreadsheet of timeline:

https://docs.google.com/spreadsheets/d/1XCx9vvBQaPycv8Q_Qt2IPNuL-2shAJdiJyBw84gj10/edit?usp=sharing

Investigator notes:

<https://docs.google.com/document/d/1Lzx70mom2cZdvwT1mzPws1-OPwfbkCrnpnG5TmaEars/edit?usp=sharing>